HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed quantitative pricing models for a range of energy contracts, including wind, solar, and battery storage structures, to assess deal-level risk across European markets.
- Analyzed power, gas, and carbon market data to inform pricing strategies and support complex deal structuring decisions for origination and trading teams.
- Built and enhanced valuation models using **Python** and **SQL**, incorporating scenario analysis and sensitivity testing for energy trading opportunities.
- Collaborated with **Origination** and **Trading** departments to support live deals, translating market insights into actionable pricing recommendations for senior stakeholders.
- Designed and implemented **cash-flow models** in **Excel** and **Python** for fixed-price PPAs and virtual products, improving valuation accuracy by approximately **18%**.
- Managed multiple competing priorities in a fast-paced trading environment, ensuring high accuracy and timely delivery of pricing analysis for over **50** distinct contract types per quarter.
- Containerized pricing and valuation microservices using **Docker** and deployed them to an **AWS** environment, improving deployment consistency and reducing setup time by **40%**.

SKILLS

Energy Markets – Pricing, Structuring, Risk Analysis, Market Analytics, Power Markets, Gas Markets, Carbon Markets

Quantitative Analysis – Valuation Techniques, Cash-Flow Modelling, Scenario Analysis, Sensitivity Analysis, Deal Structuring

Programming & Data Analysis – Python, SQL, Excel

Cloud & Containerization – AWS, Docker

Financial Products – PPAs, Battery Storage Structures, Balancing Agreements, Virtual Products

PROJECTS

Energy Contract Pricing & Risk Analysis Tool (HCL Technologies)

- Developed a **Python**-based application to automate the pricing and risk assessment of various energy contracts, integrating with market data feeds.
- Implemented **SQL** queries against a PostgreSQL database to extract and validate historical market data, ensuring consistency for pricing models.
- Designed the user interface and reporting module to present pricing recommendations and risk metrics to business analysts and stakeholders.

Portfolio Valuation Model Enhancement (MSc Data Analytics)

- Enhanced an existing valuation model using **Python** (pandas, NumPy) to incorporate advanced **scenario analysis** for a portfolio of hypothetical energy assets.
- Performed **statistical analysis** on historical market data to derive parameters for the scenario generation, improving model robustness.
- Visualized model outputs and sensitivities using **Power BI** dashboards, facilitating clearer understanding for academic review.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)